

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01 AND
EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.(BTL 3)*

Assessment Tool Weightage: 15%

Total Marks: 100 Time: 60 Mins No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Namburi Sai Yashaswini(192425250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: yashaswini

Criterion	Weight	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	

7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	

Total	100%	
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ANSWER — Question 1: Smart Healthcare Diagnosis

Scenario recap: A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.

System Overview — Knowledge Base and Inference Engine

Before mapping the logic, it helps to see how the facts and rules feed into the reasoning system. The diagnosis facts and the medical domain rules are asserted into a knowledge base; the inference engine then applies unification and resolution to answer the diagnostic query.

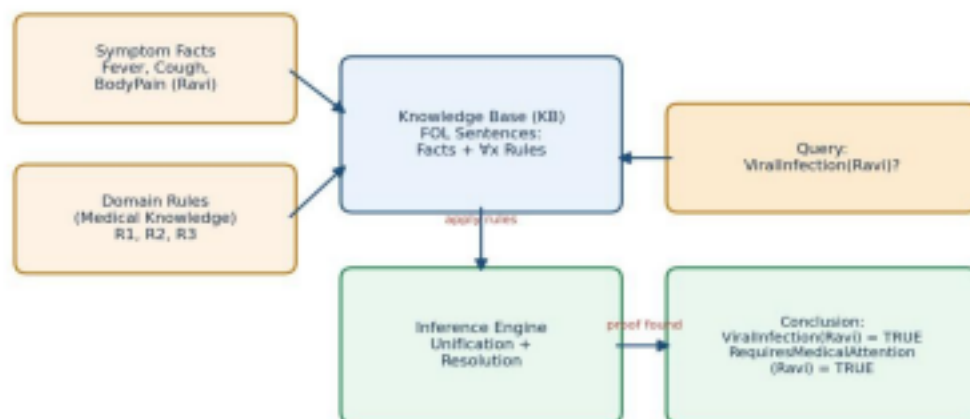


Fig. 1 — Knowledge base and inference engine architecture for Ravi's diagnosis.

(i) Facts, Entities, Predicates, Variables, and

Constants • Entity / Constant: Ravi — the specific patient under diagnosis.

• Variable: x — used to represent any generic patient in the rules.

• Given Facts: Fever(Ravi), Cough(Ravi), Body Pain(Ravi).

• Given Rules (general knowledge, true for any patient x):

Rule 1: $\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Rule 2: $\text{Flu}(x) \wedge \text{Body Pain}(x) \rightarrow \text{Viral Infection}(x)$

Rule 3: $\text{Viral Infection}(x) \rightarrow \text{Requires Medical Attention}(x)$

Table 1 lists every predicate used in this knowledge base along with its meaning and arity:

Predicate	Meaning	Arity
Fever(x)	Patient x has fever	1
Cough(x)	Patient x has a cough	1
Body Pain(x)	Patient x has body pain	1
Flu(x)	Patient x has flu	1
Viral Infection(x)	Patient x has a viral infection	1
Requires Medical Attention(x)	Patient x requires medical attention	1

(ii) Propositional Logic Representation

Since propositional logic cannot use variables or predicates, each statement is represented for the specific patient Ravi using atomic propositions:

$P = \text{Fever(Ravi)}$ $Q = \text{Cough(Ravi)}$ $R = \text{Body Pain(Ravi)}$

$S = \text{Flu(Ravi)}$ $T = \text{Viral Infection(Ravi)}$ $U = \text{Requires Medical Attention(Ravi)}$

Table 2 shows each propositional symbol with its meaning and resulting truth value once inference is complete:

Symbol	Propositional Meaning	Truth Value
P	Fever(Ravi)	True
Q	Cough(Ravi)	True
R	Body Pain(Ravi)	True

S	Flu(Ravi)	Derived — True
T	Viral Infection(Ravi)	Derived — True
U	Requires Medical Attention(Ravi)	Derived — True

Rules expressed propositionally:

$$(P \wedge Q) \rightarrow S$$

$$(S \wedge R) \rightarrow T$$

$$T \rightarrow U$$

Table 2b is a full truth table verifying the rule chain for every combination of P, Q, R. Ravi's actual case — where all three symptoms are present — is the last row, shown in bold in the discussion below.

P	Q	R	$S=(P \wedge Q)$	$T=(S \wedge R)$	$U=T$
F	F	F	F	F	F
F	F	T	F	F	F
F	T	F	F	F	F
F	T	T	F	F	F
T	F	F	F	F	F
T	F	T	F	F	F
T	T	F	T	F	F
T	T	T	T	T	T

Only the last row — Fever, Cough, and Body Pain all true (Ravi's actual case) — makes $U = \text{Requires Medical Attention(Ravi)}$ true, confirming the propositional derivation independently of the FOL/resolution proof.

(iii) First-Order Logic (FOL) Representation

FOL generalises the rules over all patients using the universal quantifier $\forall$, while facts about Ravi remain ground (constant) statements:

$$\forall x [\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)]$$

$$\forall x [\text{Flu}(x) \wedge \text{Body Pain}(x) \rightarrow \text{Viral Infection}(x)]$$

$$\forall x [\text{Viral Infection}(x) \rightarrow \text{Requires Medical Attention}(x)]$$

Facts: Fever(Ravi) , Cough(Ravi) , Body Pain(Ravi) . Goal to prove: $\text{Viral Infection(Ravi)}$?

Table 3 compares the propositional and FOL representations for this scenario:

Aspect	Propositional Logic	First-Order Logic (FOL)
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Scope		single patient (Ravi) only Generalised — patient x
Building blocks	Atomic propositions (P, Q, R, ...)	Predicates, variables, constants, quantifiers
Reusability	Must rewrite rules for every new patient	Same rule (x) applies to all patients $\forall$
Quantifiers	Not supported	Uses (universal) here $\forall$
Expressiveness	Limited — cannot express relationships expresses	generalisations

(iv) Unification and Substitution

To apply Rule 1 to Ravi's facts, the predicates Fever(x) and Cough(x) in the rule must be unified with the ground facts Fever(Ravi) and Cough(Ravi):

$\text{UNIFY}(\text{Fever}(x), \text{Fever}(\text{Ravi})) \rightarrow \text{MGU} = \{x / \text{Ravi}\}$

$\text{UNIFY}(\text{Cough}(x), \text{Cough}(\text{Ravi})) \rightarrow \text{MGU} = \{x / \text{Ravi}\}$

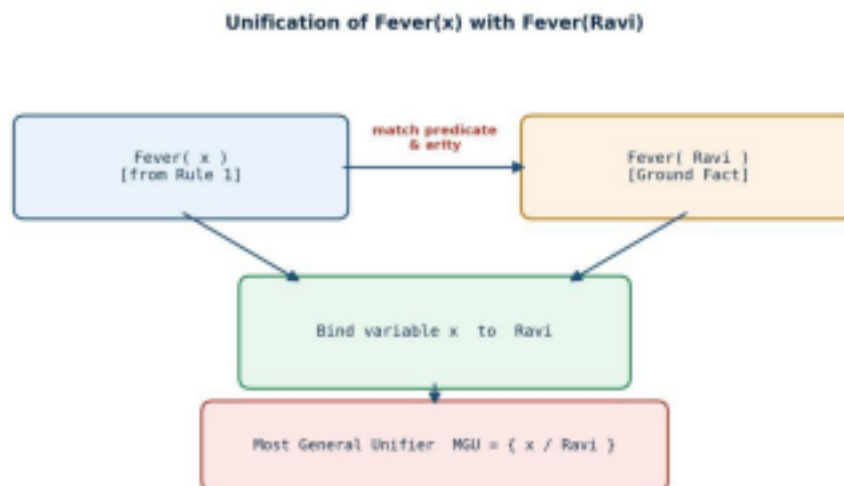


Fig. 2 —

Unification of the rule predicate Fever(x) with the ground fact Fever(Ravi).

Applying $\{x/\text{Ravi}\}$ to Rule 1 grounds it as $\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi}) \rightarrow \text{Flu}(\text{Ravi})$. The same substitution unifies every other predicate used across the three rules, summarised in Table 4:

Rule Predicate	Fact / Term	Substitution (MGU)
Fever(x)	Fever(Ravi)	$\{x / \text{Ravi}\}$
Cough(x)	Cough(Ravi)	$\{x / \text{Ravi}\}$

Flu(x)	Flu(Ravi) [derived]	{x / Ravi}
Body Pain(x)	Body Pain(Ravi)	{x / Ravi}
Viral Infection(x)	Viral Infection(Ravi) [derived]	{x / Ravi}

(v) Resolution — Proof that Ravi has a Viral

Infection Convert all rules and facts to clausal (CNF) form, and negate the goal for a proof by refutation:

C1: $\neg \text{Fever}(x) \vee \neg \text{Cough}(x) \vee \text{Flu}(x)$

C2: $\neg \text{Flu}(x) \vee \neg \text{Body Pain}(x) \vee \text{Viral Infection}(x)$

C3: $\neg \text{Viral Infection}(x) \vee \text{Requires Medical Attention}(x)$

C4: $\text{Fever}(\text{Ravi})$ C5: $\text{Cough}(\text{Ravi})$ C6: $\text{Body Pain}(\text{Ravi})$

C7 (negated goal): $\neg \text{Viral Infection}(\text{Ravi})$

Table 5 lists every resolution step, the clauses combined, the substitution applied, and the resulting resolvent clause:

Step	Clauses Resolved	Substitution	Resolvent
1	C1 $\wedge$ C4	{x/Ravi}	$\neg \text{Cough}(\text{Ravi}) \vee \text{Flu}(\text{Ravi})$
2	R1 $\wedge$ C5	—	$\text{Flu}(\text{Ravi})$
3	C2 $\wedge$ R2	{x/Ravi}	$\neg \text{Body Pain}(\text{Ravi}) \vee \text{Viral Infection}(\text{Ravi})$
4	R3 $\wedge$ C6	—	$\text{Viral Infection}(\text{Ravi})$
5	R4 $\wedge$ C7	—	NIL (empty clause)

Step 5 derives the empty clause NIL, confirming the negated goal is unsatisfiable — so $\text{Viral Infection}(\text{Ravi})$ is proved TRUE by resolution refutation. Figure 3 shows the same proof laid out as a resolution tree:

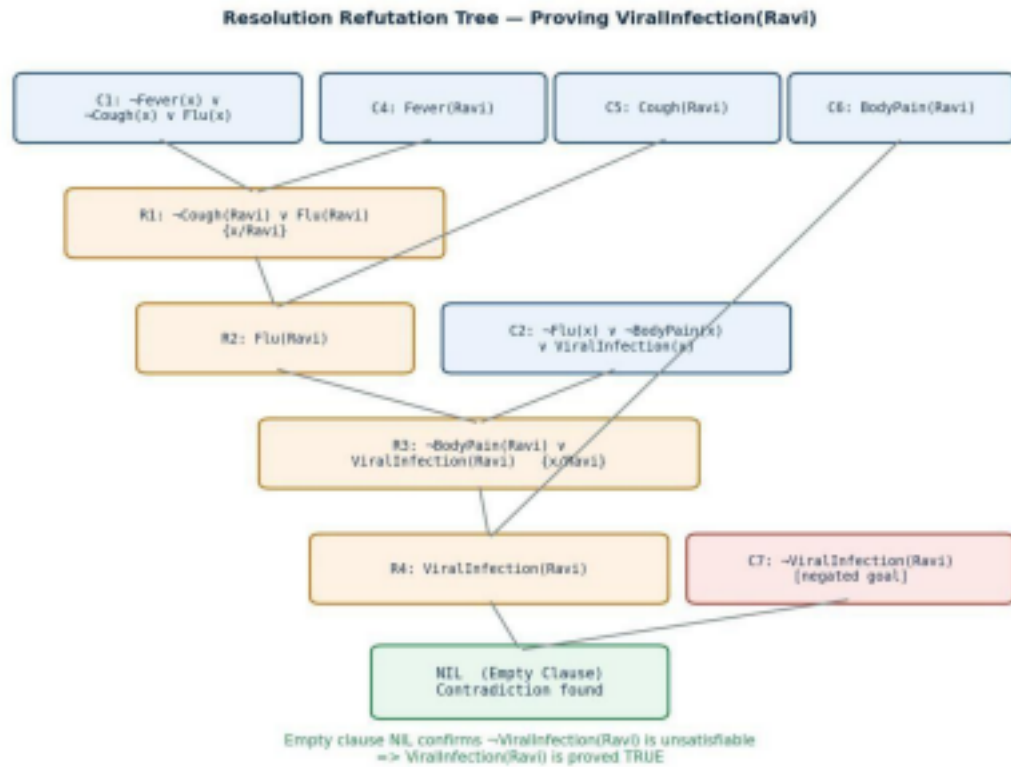


Fig. 3 — Resolution refutation tree deriving NIL from the negated goal.

(vi) Complete Inference Sequence and Final Conclusion

Table 6 walks through the forward-chaining inference sequence, from the raw facts to the final diagnostic conclusion:

Step	Applied Rule	Substitution	Fact Derived
1	Given	—	Fever(Ravi), Cough(Ravi), Body Pain(Ravi)
2	Rule 1	{x/Ravi}	Flu(Ravi)
3	Rule 2	{x/Ravi}	Viral Infection(Ravi)
4	Rule 3	{x/Ravi}	Requires Medical Attention(Ravi)

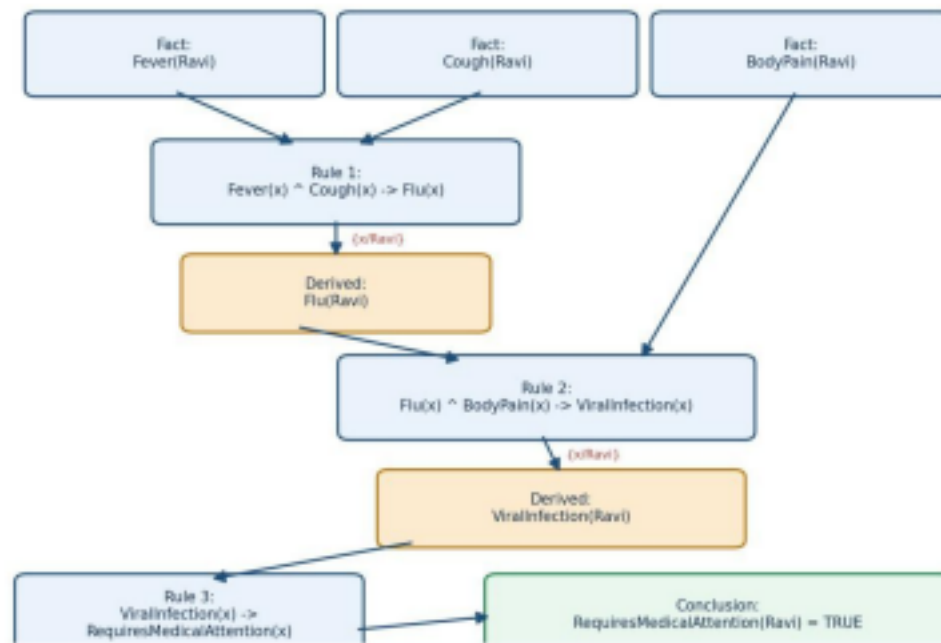


Fig. 4 — Complete mapping diagram: facts → rules → derived conclusion.

Final Conclusion: Ravi has a viral infection — Viral Infection(Ravi) = TRUE — and consequently Requires Medical Attention(Ravi) = TRUE, meaning the AI diagnosis system correctly recommends that Ravi be given medical attention.

Forward Chaining vs. Backward Chaining — Which Did We Use?

The inference sequence in Table 6 and Figure 4 is a forward-chaining derivation: it starts from known facts and fires rules until the goal is reached. The resolution proof in Table 5 and Figure 3 is effectively backward chaining by refutation: it starts from the negated goal and works backward until a contradiction is found. Table 7 compares the two strategies as applied to this diagnosis:

Aspect	Forward Chaining (Table 6)	Backward Chaining / Resolution (Table 5)
Starting point	Known facts (symptoms)	Negated goal $\neg$ Viral Infection(Ravi)
Direction	Facts → Rules → Conclusion	Goal → Clauses → Contradiction (NIL)
Best suited for	Data-driven monitoring systems	Query-answering / diagnostic systems

Result for Ravi	Requires Medical Attention(Ravi) = TRUE	NIL derived — goal proved TRUE
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Both strategies agree on the same outcome, which cross-validates the correctness of the knowledge

base and the inference steps used for Ravi's diagnosis.

Summary

- Facts about Ravi were combined with general medical rules using unification (substitution {x/Ravi}).
- Forward chaining and resolution both independently confirm Viral Infection(Ravi) = TRUE.
- The proof by refutation (Figure 3) derives the empty clause NIL, formally validating the diagnosis.
- The system's final recommendation is that Ravi requires medical attention.

EVALUATION RUBRICS:

Criterion	Weighta ge (%)	Excelle
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Problem Understanding & KnowledgeIde ntification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10- Ide mo rel kno e v mi om .
Logical Representati on & Mapping	20%	17–20% – Accurately maps real world statements into appropriate propositional and first-order logic representation s.	13- Pro mo cor log ma wit mi err

Predicates, & Variables, Constants Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
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Logical Connectives & Formula Construction	10%	9–10% –Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.
		construct valid logical formulas.			
Inference Logical Reasoning	15%	13–15% –Performs accurate stepby-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.

Unification Substitution	≤ 10%	9–10% –Correctly identifies substitutions and MostGeneral Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% –Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram Presentation	≤ 5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.

		presentation.			
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